

VALDEMAR SKOU KNUDSEN

Jemtelandsgade 6, 2. th., 2300 København S, Denmark

valdemar.skou@gmail.com • +45 42802023 • valdemarskou.github.io/

PROFESSIONAL PROFILE AND MATHEMATICAL COMPETENCIES

- MSc in mathematics; specialized in scientific computing, mathematical modelling, data analysis.
- Specific experience with using high-order orthogonal polynomials to solve partial differential equations (PDEs) and their eigenvalue problems.
- Strong theoretical knowledge in the analysis of PDEs.
- Programming proficiency in Julia, C++, Python (PyTorch, Pandas, SciPy).
- Fast learner and problem-solver; adept scientific communicator to multidisciplinary audiences.

EDUCATION

University of Copenhagen. Master of Science, mathematics (2022-2025).

Grade point average: 10,0.

Masters thesis title: “Approximating solutions to partial differential equations with physics-informed reproducing kernels”

Supervised by Oswin Krause (Associate professor, Department of Computer Science).

Notable courses: Computational Methods in Simulation, Advanced Numerical Methods for Differential Equations (Galerkin FEM), Applied Programming (C++), Mathematical Statistics, Partial Differential Equations, Differential Operators/Function Spaces.

University of Copenhagen. Bachelor of Science, mathematics (2019-2022).

Grade point average: 10,0.

Seoul Foreign High School. International Baccalaureate diploma (2013-2017). Score: 36 points.

WORK EXPERIENCE

Nørre Gymnasium

Mathematics teacher, full time (Nov 2025-)

Responsible for teaching mathematics at both the A and B level in the Danish high school system (STX). Tasks include scheduling and preparing classroom modules, creating and grading homework and larger assignments, coordinating with teachers and other staff (i.e. for multidisciplinary projects and/or general classroom cohesion), and supervising students in their large end-of-school project (the SRP project).

Department of Mathematical Sciences, University of Copenhagen

Teaching assistant, part time (2021-2024):

Responsible for organizing and holding exercise classes for university level mathematics courses, with multidisciplinary audiences (students from biology, physics, CS, mathematics). Tasks include presenting solutions to the classroom, trouble-shooting with students (i.e. software issues), grading assignments; working in tandem with the lecturer in order to ensure that the course runs effectively.

Technical University of Denmark (DTU)

Student assistant in the Office for Research and Relations, part time (2017-2019):

Responsible for various tasks to support the office, including expense settlements, collecting and cleaning data on DTUs joint publications and collaborations with other universities; publishing news articles. Required knowledge of various DTU related systems and software.

PERSONAL INFORMATION

- Amateur cellist; I was section principal in a prominent youth orchestra, and later in an APAC orchestra. Nowadays I play a lot of solo repertoire and traditional Scandinavian folk music.
- In my spare time I continue to tinker with scientific computing, mostly learning about PDE modelling and simulation methods that I implement in a high-level language (usually Julia or Python). I sometimes journal this progress on my personal website.
- International experience and background; I attended high-school in South Korea, where I met and studied with people from all over the world. I never learned Korean, but remain fluent in Danish and English.